Earth magnetic field navigation algorithms for aerospace vehicles

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Problem Statement

Application:

- High-altitude map-less navigation using Earth's ambient magnetic field

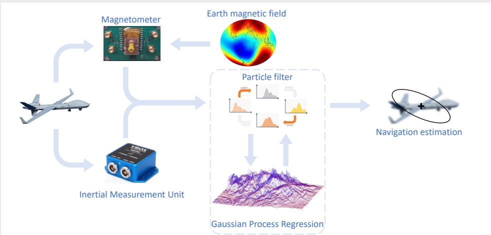

Figure 1: Inertial and magnetic data fusion

State estimation:

- Inertial navigation, based on the integration of noisy inputs from inertial units, yields a solution which drifts over time
- Data fusion from multiple sensors helps reduce this drift
- GNSS data can be blurred or unavailable
- SLAM approach using surrounding physical fields (magnetic, gravimetric, altimetric, ...)

Regularised Particle Filter (RPF)

Particle filtering [1]:

- Recursive Bayesian filtering based on Monte-Carlo methods
- Able to tackle highly non-linear, ambiguous and multimodal problems
- State estimation:

$$\mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}(\varphi(\mathbf{X}_{0:k})) \approx \sum_{i=1}^{N_s} w_k^{(i)} \, (\delta * \varphi) \left(\mathbf{x}_{0:k}^{(i)}\right)$$

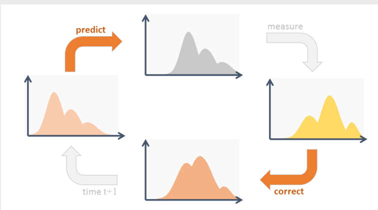

Figure 2: Usual state estimation scheme

RPF [2]:

- Particle degeneracy and impoverishment are two common problems with particle filters
- Regularisation of the PDF using kernel mixtures to solve these issues:

$$\mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}(\varphi(\mathbf{X}_{0:k})) \approx \sum_{i=1}^{N_s} w_k^{(i)} \, (\mathcal{K}_h * \varphi) \left(\mathbf{x}_{0:k}^{(i)}\right)$$

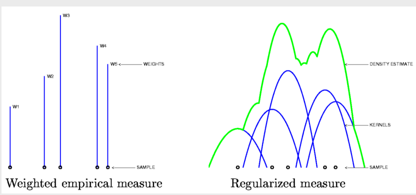

Figure 3: Barycentric Dirac and kernel approximations of the PDF

Gaussian Process Regression (GPR)

Terrain-aided navigation (TAN):

- In TAN situations, the altimetric/bathymetric field is usually known
- The PDF of the state can be computed by comparing local measures of the field to the known map
- If no map is given, SLAM methods can be used to build one on the fly

Kriging [3]:

- GPR can be used to create a map using simple prior knowledge of the field or sparse noisy data, which can be used by the RPF

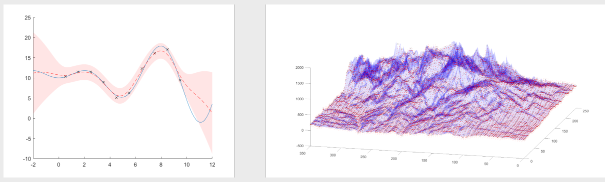

Figure 4: Function and map interpolations using GPR

Numerical Results

Experimental setup:

- TAN with known altimetric field (unless stated otherwise)
- 1000 Monte-Carlo realisations per filter
- 500 particles per realisation

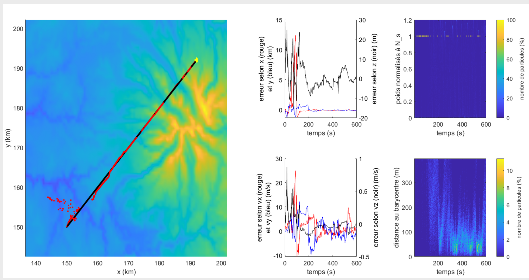

Figure 5: Successful trajectory estimation with a RPF

Results:

Tested filter	Convergence rate	final RMSE
SIS	13.8 %	0.67 km
SIR	16.6 %	0.38 km
PF	26.9 %	0.34 km
RPF	84.7 %	0.20 km
RPF + kriging	96.7 %	0.36 km

Table 1: Metrics for the assessed filters

References

[1] M. S. Arulampalam, S. Maskell, N. Gordon and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking", *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 174-188, Feb. 2002

[2] C. Musso, N. Oudjane, and F. Le Gland, "Improving regularised particle filters", *Sequential Monte Carlo methods in practice*, New York, NY: Springer New York, 2001. 247-271

[3] C. Palmier, A. Giremus, P. Minvielle, C. Vacar and G. Bourmaud, "Terrain-Aided SLAM with Limited-Size Reference Maps Using Gaussian Processes", *2023 31st European Signal Processing Conference (EUSIPCO)*, Helsinki, Finland, 2023, pp. 830-834